IMAN SHARIFI

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Objective

As a diligent, determined, and self-motivated researcher, my interests lie at the intersection of Robotics, Control, Machine Learning, and Computer Vision. I have a strong passion for using cutting-edge AI algorithms like Neuro-Symbolic Reinforcement Learning in the decision-making and Control phase of Cyber-Physical Systems like Autonomous Driving, and use innovative ideas in Computer Vision for the perception phase.

Education

- **Master of Science, Mechanical Engineering** 2019 - 2021
  School of Mechanical Engineering, College of Engineering, Sharif University of Technology
  GPA: 18.62/20
- **Bachelor of Science, Aerospace Engineering** 2015 - 2019
  School of Aerospace Engineering, College of Engineering, K. N. Toosi University of Technology
  GPA: 15.28/20
- **Diploma of Education, Mathematics and Physics** 2011 - 2015
  Shahid Sani Government Model School
  GPA: 19.43/20

Experience

- **Research Assistant Intern** Nov. 2022 - Present
  University of Surrey, Department of Mechanical Engineering
  CAV-Lab (Connected & Automated Vehicles Lab)
  Research Area: Neuro-Symbolic Reinforcement Learning using Inductive Logic Programming
  +Remote AI Research Assistant Intern
- **Graduate Research Assistant** 2019 - 2022
  Sharif University of Technology, School of Mechanical Engineering
  Research area: Application of neural-network-based reinforcement learning in robotics, especially quadcopter
  Supervisor: Prof. A. Alasty
- **Graduate Teaching Assistant, Fuzzy control** 2022 - 2022
  Sharif University of Technology, School of Mechanical Engineering
  Under the supervision of Prof. A. Alasty
- **Graduate Teaching Assistant, Advanced Mathematics** 2019 - 2020
  Sharif University of Technology, School of Mechanical Engineering
  Under the supervision of Dr. Alireza Taheri
- **Undergraduate Teaching Assistant, Automatic Control** 2019 - 2020
  Sharif University of Technology, School of Mechanical Engineering
  Under the supervision of Prof. H. Salarieh
- **Undergraduate Research Assistant** 2017 - 2019
  K. N. Toosi University of Technology, School of Aerospace Engineering
  Research area: Dynamic and Control of robots, particularly Mobile Robots
  Supervisor: Dr. A. B. Novinzadeh
- **Undergraduate Teaching Assistant, Differential Equations** 2017 - 2018
  K. N. Toosi University of Technology, School of Aerospace Engineering
  Under the supervision of Dr. M. Jafari Nodoshan
- **SONEX (Sharif Ultrasonic Company)** 2020 - 2021
  Research and Design (R & D)
  + Research and Design Non-destructive Testing (NDT) Transducers

+ Teamwork experience

Skills

- Machine Learning: RL, SK-learn, Keras, PyTorch, Google Colab, CNN, RNN
- Symbolic Programming: Inductive Logic Programming (ILP) using Prolog, Aleph, Metagol
- Computer Vision: OpenCV, CNN
- Dynamics and Control: MATLAB, SIMULINK, ROS
- Programming: C, Python, Prolog, Java
- Computer-Aided Design: Solidworks, CATIA
- Mechatronics: Arduino, PLC
- Writing Essay: LaTeX, Microsoft Office
- Web Design: HTML, CSS

Achievements & Awards

- Top Bachelor's Thesis of Iranian Aerospace Association
- Among the first 0.1% accepted students in the Master's Entrance Exam of Mechanical Engineering among about 15,000 people
- Selected student for Ahmadi Roshan project of the National Elite Foundation of Iran
- Among the first 3% accepted students in the Iranian Nationwide Entrance Exam of Mathematics and Physics among about 200,000 participants

Courses

- Nonlinear Control, 19.7/20 | Advanced Mathematics, 19.4/20 | Computer Vision, 19.2/20 | Automatic Control, 19.0/20 | Advanced Dynamics, 18.5/20 | Advanced Control, 18.2/20 | Fuzzy Control, 18.1/20 | Control Systems Design, 17.0/20
- Machine Learning, Artificial intelligence, Deep Learning (Audited) | Reinforcemet Learning and Adaptive Control (Audited) | Model Predictive Control (MPC) and Optimal Control
- Flight Dynamics and Control | Guidance and Navigation

Projects

- **Neuro-Symbolic Reinforcement Learning using Inductive Logic Programming for Autonomous Highway Driving**
  Ongoing project, under the supervision of Dr. Fallah
- **Safe Reinforcement Learning using Symbolic Logical Programming for Autonomous Highway Driving**
  Under the supervision of Dr. Fallah
- **Safe Symbolic Q-Learning using Inductive Logic Programming in Maze Grid-World**
  Under the Supervision of Dr. Fallah
- **Design Self-Tuning PID Control via a Hybrid Neural Structure based on Reinforcement Learning for Quadcopter Attitude and Altitude Control**
  Master's Thesis, under the supervision of Prof. A. Alasty
- **Trajectory Tracking of Tiltrotor using Fuzzy Sliding Mode Control with Fuzzy Modeling of Dynamic**
  Fuzzy Control project, under the supervision of Prof. Aria Alasty
- **Quadcopter Trajectory Tracking using Adaptive Nonlinear Algorithms (LQR, Sliding Mode, Backstepping, Feedback Linearization, Model Reference Adaptive Control)**
  Nonlinear Control project, under the supervision of Prof. Aria Alasty
- **Digits Classifications and Localization via Convolutional Neural Networks (CNN) and Face Detection using Histogram of Oriented Gradients (HOG) Algorithm**
  Computer Vision project, under the supervision of Dr. Hoda Mohammadzadeh
- **Video Synopsis in a Video File Recorded by a Surveillance Camera**
  Computer Vision project, under the supervision of Dr. Hoda Mohammadzadeh
- **Simulation of a 3-DOF Manipulator on Satellite Body using Newtonian, Lagrangian, and Quaternion Method**

Advanced Dynamics project, under the supervision of Dr. Hossein Nejat

- **Modeling, Simulation, Design, and Fabrication of Two-Wheel Mobile Balanced Robot (2WMBR)**
  Bachelor's Thesis, under the supervision of Dr. Alireza B. Novinzadeh

Publications

- **Design Self-Tuning PID Control via a Hybrid Neural Structure based on Actor-Critic Method for Quadcopter Attitude and Altitude Control**
  I. Sharifi, A. Alasty
  30th Annual International Conference of Iranian Society of Mechanical Engineers (ISME 2022), May 25-26, Tehran, Iran.
- **Safe Reinforcement Learning via Symbolic Logical Programming for Autonomous Highway Driving**
  I. Sharifi, S. Fallah
  (In preparation)

Languages

- English, Fluent
- Official TOEFL iBT certificate, Overall score: 96/120. Reading: 25/30, Listening: 24/30, Speaking: 23/30, Writing: 24/30

Personal Details

- Date of Birth : 12/10/1996
- GitHub : https://github.com/iman-sharifi-ghb?tab=repositories

Reference

- **Saber Fallah - University of Surrey, Guildford, UK**
  Associate Professor
  s.fallah@surrey.ac.uk
- **Aria Alasty - Sharif University of Technology, Tehran, Iran**
  Full Professor
  aalasti@sharif.edu
- **Alireza Taheri - Sharif University of Technology, Tehran, Iran**
  Assistant Professor
  artaheri@sharif.edu
- **Alireza B. S. Novinzadeh - K. N. Toosi University of Technology, Tehran, Iran**
  Associate Professor
  Novinzade@email.kntu.ac.ir
- **Mahdi Jafari Nodooshan - K. N. Toosi University of Technology**
  Assistant Professor
  Jafari@email.kntu.ac.ir